

Science 7, Module 3: Metals and Their Properties

Quarter 1. A sample module written for the QuizApp demo.

What you will learn. After this module you can describe the physical properties of metals, explain why metals conduct electricity, tell metals from nonmetals, and name common alloys and their uses.

What are metals?

Most of the elements in the periodic table are metals. Metals are usually shiny, hard and dense, and most of them are solid at room temperature. Mercury is the only metal that is liquid at room temperature. It melts at about -38.8°C , which is why it was once used in thermometers.

Gallium and cesium are solid at room temperature, but they melt just above it. Gallium melts at about 29.8°C , so a piece of gallium melts in the hand.

Physical properties of metals

Luster. Metals reflect light, so a freshly cut or polished metal surface is shiny. This property is called luster.

Malleability. Metals can be hammered or pressed into thin sheets without breaking. Aluminum foil is made this way. This property is called malleability.

Ductility. Metals can be drawn into thin wires. Copper is used for electrical wiring because it is both ductile and a good conductor. This property is called ductility.

Conductivity. Metals are good conductors of heat and electricity. The electrons in a metal are free to move, which lets an electric current flow through it. Silver is the best conductor of electricity, but copper is used more often because it costs much less.

Metals and nonmetals

Property	Metals	Nonmetals
Appearance	Shiny (luster)	Dull
State at room temperature	Mostly solid; mercury is liquid	Solid, liquid or gas; bromine is liquid
Conductivity	Good conductors of heat and electricity	Poor conductors, except graphite
When hammered	Flatten into sheets	Break or crumble (brittle)

Bromine is the only nonmetal that is liquid at room temperature. Like mercury, it is a liquid, but it is not a metal.

Corrosion

Many metals react with oxygen and water in the air. When iron reacts with oxygen and water, it forms iron oxide, which we call rust. This slow wearing away of a metal by chemical reaction is called corrosion. Painting, oiling or coating iron with zinc protects it from rusting. Coating iron with zinc is called galvanizing.

Alloys

An alloy is a mixture of a metal with one or more other elements. Alloys are made because they are often stronger or more useful than the pure metal.

Alloy	Made of	Common use
Steel	Iron and a small amount of carbon	Buildings, bridges, tools
Stainless steel	Iron, chromium and nickel	Kitchen utensils; it does not rust easily
Brass	Copper and zinc	Musical instruments, door knobs
Bronze	Copper and tin	Medals, statues, bells

Metals in the Philippines

The Philippines is one of the world's largest producers of nickel. Much of the country's nickel ore comes from mines in Surigao and Palawan. The country also mines gold and copper, mainly in the Cordillera region and in Mindanao.

Mining brings jobs and income, but it can also damage forests, rivers and farmland. Responsible mining includes replanting mined areas and treating the water that leaves a mine.

Safety

Mercury is toxic. Its vapor can damage the brain and kidneys, so a broken mercury thermometer should never be cleaned up with bare hands or a vacuum cleaner. Today, most thermometers use alcohol or a digital sensor instead of mercury.

Key words

Word	Meaning
Luster	The shine of a metal surface
Malleability	Being able to be hammered into sheets
Ductility	Being able to be drawn into wires
Corrosion	The wearing away of a metal by chemical reaction
Alloy	A mixture of a metal with other elements

Written for the QuizApp demo (RAITE 2026). Free to use for testing.